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17 November 1972

Superseding
XWS-12820(USW)
23 November 1971

MILITARY SPECIFICATION

CABLE, ELECTRIC, MISSILE, 54-CONDUCTOR

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. This specification covers the minimum requirements for the procurement of a synthetic rubber-jacketed cable assembly embodying 54 conductors. The cable assembly is used in a missile fire control system.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-I-631	Insulation, Electrical, Synthetic-Resin Composition, Nonrigid
MIL-C-915	Cable, Electrical, Special Purpose; General Specification for
MIL-Y-1140	Yarn, Cord, Sleeving, Cloth and Tape-Glass
MIL-R-6855	Rubber, Synthetic, Sheets, Strips Molded or Extruded Shapes
MIL-W-16878	Wire, Electrical, Insulated, High Temperature
MIL-C-19787	Cable, Electrical, Torpedo, 65 Conductor (for Torpedo Control Electric Setting)

FSC 1355

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STANDARDS

Federal

Federal Test Method Standard No. 228	Cable and Wire, Insulated, Methods of Testing

DRAWINGS

Naval Ordnance Systems Command

LD 162872	Cable, Torpedo Tube Missile Control, Mk 8 Mod 0
LD 162873	Cable, Torpedo Tube Missile Control, Mk 9 Mod 0

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The multiconductor cable covered by this specification is an extra flexible, insulated, concentric control cable having an abrasive-resistant jacket with the external diameter held to close tolerances. The cable is suitable for continuous service in a submarine environment.

3.2 General. Electric cable shall be of the design shown in Figure 1, and shall conform to the requirements of this specification and the contract or order.

3.2.1 Preproduction sample. Unless otherwise specified in the contract or order (see 6.2), a preproduction sample of electric cable is required. The sample shall be manufactured using the methods and procedures proposed for the production lot. This sample will be tested as specified in 4.3.2 and is for the purpose of determining, prior to starting production, that the manufacturer's production process will produce cables that will meet the requirements of this specification.

3.2.2 Manufactured length. Unless otherwise specified in the contract or order, the manufactured length shall consist of any one single length of cable to which the impervious jacket was applied in one length.

3.2.3 Conflicting requirements. In the event of conflict between the documents referenced herein and the contents of this specification, the contents of this specification shall apply.

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3.3 Design and construction

3.3.1 Materials. All materials used in construction of the cable shall be of a quality and form which will assure compliance with the requirements of this specification and applicable drawings.

3.3.2 Conductors. All conductors shall conform to MIL-W-16878 and shall be of extruded teflon insulated in accordance with that specification. Wire AWG #18 shall be Type E, 19/30 strand and wire AWG #12 shall be Type EE, 19/25 strand. The stranded wire shall be of silver-plated copper. The number, shielding, and size of conductors shall be as specified in Table I of this specification. The wire insulation shall be color coded as specified also in Table I.

3.3.3 Conductor assembly. Assembly of the conductors shall be in accordance with Table II.

3.3.4 Cable sheath. As shown in Figure 1, the sheath surrounding the conductor assembly shall be concentrically applied and shall consist of tough, resilient, abrasion-resistant polychloroprene or polyurethane. The sheath shall be homogeneous in character, free from porosity, of a quality typical of best manufacturing procedures, and shall conform to MIL-R-6855, Class 2, Grade 60. In addition, the material used shall be compatible for potting and/or bonding, with the surfaces being free of residual release agents. The cable sheath shall have an average minimum thickness of 0.94 inch and a minimum thickness of 0.80 inch.

3.3.5 Shielding. The conductors shall be shielded, as specified in Table I, in accordance with MIL-W-16878, except that there shall be no jackets over the shields. Shields are to be isolated from their insulated conductors by a polyethylene terephthalate film, MIL-I-631, Type G, Form F. All shields of shielded conductors are to be in intimate contact over the entire length of cable. The lay shall give maximum flexibility.

3.3.6 Filler. Glass fiber filler in accordance with MIL-Y-1140 shall be used as required to make a compact circular cross section.

3.3.7 End seals. The end seals of the cable shall be as specified in MIL-C-915. Approved alternate seals may be used if necessary.

3.3.8 Dimensions and tolerances. The overall dimensions of the cable and the jacket wall thicknesses shall conform to dimensions shown in Figure 1.

3.3.9 Wire splices. No wire splices, broken wires, broken strands of wire, or other discontinuities shall be permitted in a cable length.

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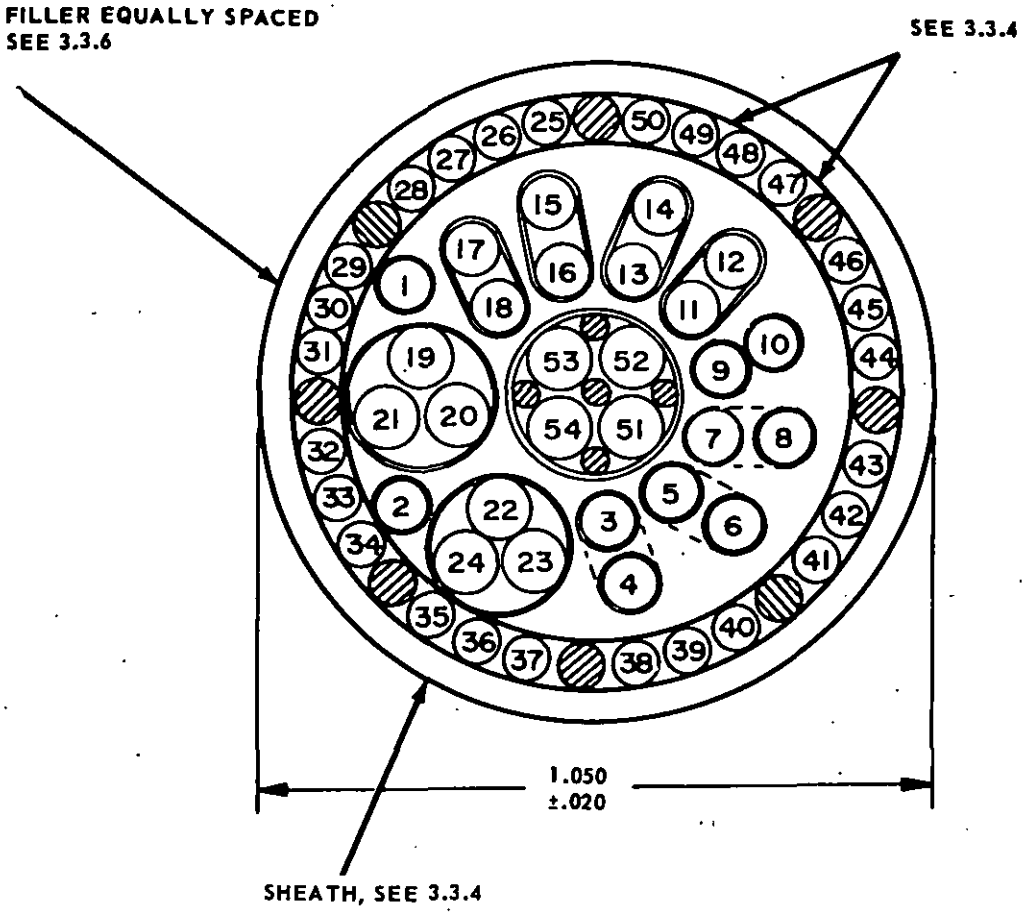


FIGURE 1. ELECTRIC CABLE CROSS-SECTION.

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3.4 Electrical requirements. The cable shall successfully meet the following electrical requirements:

3.4.1 Dielectric strength. All lengths of completed cable shall withstand, without failure, a 5-second application of a 1000-volt ac test potential, applied to each conductor sequentially with all adjacent conductors and shields at ground potential, when tested in accordance with 4.4.3.1.

3.4.2 Insulation resistance. Each length of cable shall show an insulation resistance of at least 100 megohms for 1000 feet with 500 to 750 volts dc applied to each conductor sequentially with all conductors and shields at ground potential when tested in accordance with 4.4.3.2.

3.4.3 Capacitance. The capacitance of each shielded lead in the cable with respect to its shield shall be not less than 55 mmf nor more than 90 mmf per foot, when measured at a frequency of 1 kiloHertz (kHz) ±20 Hz and tested in accordance with 4.4.3.3.

3.5 Mechanical requirements. The cable shall be capable of meeting the following mechanical requirements:

3.5.1 Cold temperature bend. The cable sheath, conductor insulation, and conductors shall show no evidence of failure when the completed cable is subjected to the cold temperature 90-degree bend test of 4.4.4.1.

3.5.2 Cable strength. The cable sheath, conductor insulation, and conductors shall show no evidence of failure when the cable is tested in accordance with 4.4.4.2.

3.5.3 Torsion. The cable sheath, conductor insulation, and conductors shall show no evidence of failure when the completed cable is tested in accordance with 4.4.4.3.

3.6 Environmental requirements. The cable shall be capable of meeting the other requirements of this specification after being subjected to the following environmental conditions:

3.6.1 Salt water immersion. The cable sheath, conductor insulation, and conductors shall show no evidence of failure when the completed cable is tested in accordance with 4.4.5.1.

3.6.2 Temperature and humidity. The cable sheath, conductor insulation, and conductors shall show no evidence of failure when the completed cable is tested in accordance with 4.4.5.2.

3.6.3 Temperature and humidity (storage mode). The cable sheath, conductor insulation, and conductors shall show no evidence of failure when the completed cable is tested in accordance with 4.4.5.3.

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TABLE I

CONDUCTOR			COLOR CODE		
CONDUCTOR NUMBER	SHIELDING REQUIREMENTS	AWG NO.	BASE COLOR	WIDE TRACER	NARROW TRACER
1	Shielded	18	Black	None	None
2	Shielded		Brown		
3	2/C Shielded & Twisted		Red		
4			Orange		
5	2/C Shielded & Twisted		Yellow		
6			Green		
7	2/C Shielded & Twisted		Blue		
8			Violet		
9	Shielded		Gray		
10	Shielded		White	Black	Brown
11	2/C Shielded & Twisted				Red
12					Orange
13	↕				Yellow
14					Green
15					Blue
16					Violet
17	2/C Twisted & Shielded				Gray
18					Black-2
19	3/C Twisted & Shielded			Brown	Black
20					Brown
21					Red
22	3/C Twisted & Shielded				Orange
23					Yellow
24					Green
25	None				Blue
26					Violet
27					Gray
28					Black-2
29				Red	Black
30					Brown
31					Red
32					Orange
33					Yellow
34					Green
35					Blue
36					Violet
37					Gray
38					Black-2
39				Orange	Black
40					Brown
41					Red
42					Orange
43					Yellow
44					Green
45					Blue
46					Violet
47					Gray
48					Black-2
49				Yellow	Black
50					Brown
51	4/Cond. With Filler for Circular Sec. & Shielded	12			Red
52					Orange
53					Yellow
54					

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TABLE II - CONDUCTOR ASSEMBLY

LAYER	CONDUCTORS	SEQUENCE	LENGTH OF LAY
Core	Filler	-	-
1st Layer	4	(51, 52, 53, 54)	Max 3.5 in. L.H.
2nd Layer	24	1, (19, 20, 21), 2, (22, 23, 24), (3, 4), (5, 6), (7, 8), 9, 10, (11, 12), (13, 14), (15, 16), (17, 18)	Max 8.0 in.R.H.
3rd Layer	26	25 thru 50	Max 9.0 in. L.H.

3.6.4 Tropical environment. The cable sheath, conductor insulation, and conductors shall show no evidence of failure when the completed cable is tested in accordance with 4.4.5.4.

3.7 Workmanship. The workmanship shall be first-class in every respect and uniform throughout each lot of cable offered for delivery. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling

4.2.1 Lot. A lot shall be defined as follows, based on footage ordered:

Order	Lot
Up to 10,000 feet	Total order
10,000 to 50,000 feet	10,000 feet
Over 50,000 feet	25,000 feet

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4.2.2 Number of samples

4.2.2.1 Preproduction. Unless otherwise specified herein or in the contract or order (see 6.2), a 100-foot preproduction sample of cable shall be supplied by the contractor. The cable shall be manufactured using the same procedures and methods and at the same plant location proposed by contractor for the execution of the contract. The sample shall be submitted to the procuring activity for purposes of determining compliance with the requirements of the contract or order and the applicable specifications. Further production of the item by the contractor prior to approval of the procuring activity or completion of inspection and tests on the preproduction sample shall be at the contractor's risk. Accepted preproduction samples will become the property of the procuring activity and will be included in the quantity of cable called for in the contract or order.

4.2.2.2 Lot acceptance. Unless otherwise specified herein or in the contract or order (see 6.2), sampling for lot acceptance shall be conducted in accordance with MIL-C-915.

4.3 Inspection and tests

4.3.1 Test conditions. Unless otherwise specified, tests and examinations required by this specification shall be conducted under any combination of conditions within the ranges below. Any specified condition shall not affect the other two ambient ranges.

(a) Temperature: 20 degrees C to 30 degrees C.

(b) Relative Humidity: Room ambient to 95 percent.

(c) Barometric Pressure: 24 to 31 inches of mercury.

4.3.2 Preproduction testing and lot acceptance testing. Preproduction testing and lot acceptance testing shall consist of the inspection and tests shown in Table III on the samples selected under 4.2.2.1 and 4.2.2.2.

4.3.3 Preproduction test waiver. If a contractor has previously produced and delivered cable in accordance with the requirements of this specification and his product has proven satisfactory, and if no change has been made in the specifications or methods of manufacture since the original test, preproduction tests for subsequent contracts or orders will be required only if specified in the contract by the procuring activity.

4.3.4 Action in case of failure

4.3.4.1 Preproduction sample. In case of failure in any preproduction sample, the cause shall be determined and corrected. A statement of the

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failure and proposed corrective action shall be submitted to the procuring activity for approval before additional tests are conducted.

TABLE III - PREPRODUCTION AND LOT ACCEPTANCE TESTS

Type Test	Test Title	Requirement	Method
Preprod/Acceptance	Visual and Dimensional Examination	3.2, 3.3 3.7	4.4.1.1
Preproduction	Cable Sheath	3.3.4	4.4.2.1
Preprod/Acceptance	Dielectric Strength	3.4.1	4.4.3.1
Preprod/Acceptance	Insulation Resistance	3.4.2	4.4.3.2
Preproduction	Capacitance	3.4.3	4.4.3.3
	Cold Temperature Bend	3.5.1	4.4.4.1
	Cable Strength	3.5.2	4.4.4.2
	Torsion	3.5.3	4.4.4.3
	Salt Water Immersion	3.6.1	4.4.5.1
	Temperature and Humidity	3.6.2	4.4.5.2
	Temperature and Humidity (storage mode)	3.6.3	4.4.5.3
	Tropical Environment	3.6.4	4.4.5.4

4.3.4.2 Lot acceptance sample. In case of failure in lot acceptance tests, the cause shall be determined and a statement of the failure and proposed corrective action shall be submitted to the procuring activity for consideration and approval. No subsequent testing shall be conducted without the approval of the procuring activity.

If corrective action is approved, the lot is to be subjected to all acceptance tests of 4.3.2 unless repetition of tests which have already been conducted is waived by the procuring activity. If corrective action is not

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approved, the lot shall be rejected. Portions of a rejected lot may be subject to retest and subsequent acceptance at the discretion of the procuring activity.

4.4 Test methods

4.4.1 General. The methods employed for testing component materials and cable performance shall be as specified in MIL-C-19787, except as modifed or supplemented herein.

4.4.1.1 Visual and dimensional inspection. The sample length of cable selected under 4.2.2 shall be examined visually and dimensionally for compliance with the requirements of this specification and other applicable specifications. In addition, each manufactured length of cable shall be examined visually for surface flaws such as pin holes and pushbacks in the sheath and shall be cut at any such defects.

4.4.2 Design and construction

4.4.2.1 Cable sheath. One sheath specimen shall be taken from each of the cable samples chosen for insulation testing for physical tests to determine compliance with MIL-R-6855 and the requirements of 3.3.4.

4.4.3 Electrical

4.4.3.1 Dielectric strength. A length of completed cable shall be tested dry, without immersion in water, in accordance with Federal Test Method Std. No. 228 (Method 6111) to determine conformance with 3.4.1.

4.4.3.2 Insulation resistance. A length of completed cable shall be tested dry, without immersion in water, in accordance with Federal Test Method Std. No. 228 (Method 6031) to determine conformance with 3.4.2.

4.4.3.3 Capacitance. The samples used shall be prepared in accordance with 4.4.3.3.1 and shall be 20 feet in length. The measurement technique employed shall give results that are reproducible to three significant figures.

4.4.3.3.1 Preparation for tests. Prepare each 20 foot minimum length of cable as follows (see Figure 2):

(a) Strip back cable sheathing 1½ to 2 inches on each end of sample for connections.

(b) Comb out shielding on each of the seven shielded conductors to within approximately ½ inch of the shielding.

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(c) Strip approximately ½ inch of insulation from each shielded conductor.

NOTE: For computation purposes, the length of the cable shall be considered as the distance between the undisturbed ends of the shielding.

4.4.3.3.2 A length of prepared cable shall be tested to determine conformance with 3.4.3.

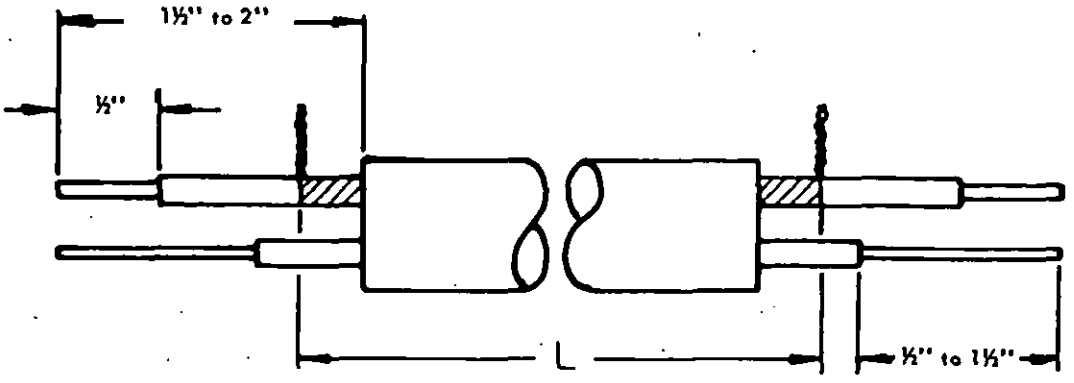


FIGURE 2. PREPARATION OF SAMPLE FOR CAPACITANCE TEST.

4.4.4 Mechanical

4.4.4.1 Cold temperature (90-degree bend test). While the cable is mounted in a suitable fixture, a 90-degree bend test shall be performed at -40 ±3 degrees F. The cable assembly and test fixture shall be conditioned at the test temperature for at least 12 hours prior to initiation of the test.

The cable shall be mounted and clamped at two points on the test fixture such that a 90-degree bend with a radius not to exceed 6 inches shall be formed within a 12-inch length of cable (see Figure 3). The cable shall be rigidly secured to the underside of a fixed support lying in a horizontal

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plane at one clamp position, and rigidly secured to a pivoting arm which shall be so mounted as to oscillate from a vertical position to a horizontal position. When in its horizontal position, its axis shall form an extension of the plane of the fixed support, and the cable shall be attached to the underside of the arm. The attachment of the cable to the oscillating arm shall be made when the arm is in the vertical position. The pivot axis of the oscillating arm shall intersect the longitudinal axis of the fixed support and shall be equidistant from the two clamping points. The fixture shall be so designed that the section of cable between the clamp positions shall not touch any portion of the mechanism or cold box during the test.

One cycle of test shall consist of an oscillation of the pivot arm from the vertical position to the horizontal and back to the vertical. The test shall be conducted at a rate of one cycle per minute with each half cycle to be accomplished within 6 seconds. The delay between cycles may occur at the end of either half cycle or may be divided to occur at the end of each half cycle.

Continuity of all conductors shall be monitored continuously. The test shall be stopped and the number of cycles recorded in the event of failure of any conductor. The test shall consist of 500 cycles of bending. At the conclusion of this test the cable must meet the requirements of 3.4.2.

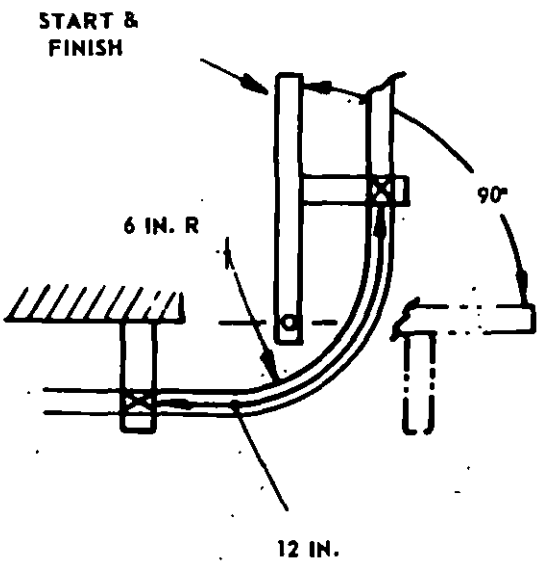


FIGURE 3. TEST SETUP-COLD TEMPERATURE BEND.

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4.4.4.2 Cable strength. The cable shall be securely clamped at two points along its length 15 inches apart. The equipment shall be oriented such that the axis of the 15-inch length of cable is in a horizontal plane. At a point midway between the clamps, a weight of 200 pounds shall be suspended from the cable for a period of at least 60 seconds, not including the application and removal periods. The shoe or weight support shall be no more than 1½ inches wide and of such design as to spread the 200-pound load over the 1½ inch length of cable. The cable sheath may be protected or padded to spread the load and prevent damage to the cable sheath. The load shall be applied and released a total of 20 times over a period not to exceed 2 hours. At the conclusion of the test, all conductors in the cable shall evidence continuity and the requirements of 3.4.2 must be met.

4.4.4.3 Torsion test. While it is mounted in a suitable test fixture, a torsion test shall be performed on the cable assembly. The cable shall be mounted and clamped at two points on the test fixture such that a 90-degree bend with a radius not to exceed 4 inches shall be formed within a 10-inch length of cable (see Figure 4). The cable shall be rigidly secured to a fixed support at one clamp position, and rigidly secured to a pivot arm at the second clamp position. The pivot axis of the arm shall be an extension of the centerline of the cable which is clamped to the fixed support.

One cycle of torsion shall consist of a 90-degree rotation in one direction, a 180-degree rotation in the reverse direction, and a 90-degree rotation in the original direction to the start position, in a plane perpendicular to the fixed support.

A test shall be conducted at a rate of 2 cycles per minute. The time duration of one cycle shall approximate but not exceed 15 seconds. Continuity of all conductors shall be monitored continuously. The test shall be stopped and the number of cycles recorded in the event of failure of any conductor. The test shall consist of 3000 cycles of torsion. At the conclusion of this test the requirements of 3.4.2 must be met.

4.4.5 Environmental

4.4.5.1 Salt water immersion. A length of completed cable shall be subjected to a 30-day immersion test. The cable shall be attached to a fixture so that the exposed ends will not be immersed in sea water. The cable shall be immersed for 16 hours and removed for a period of 8 hours. This sequence shall be repeated 30 times. No fresh water rinse will be permitted during the 30 cycles. At the conclusion of this test the cable will be required to comply with the requirements of 3.4.2.

4.4.5.2 Temperature and humidity. A 20-foot length of completed cable shall be placed in a test chamber such that the exposed ends are not inside the chamber and are prepared for monitoring the electrical characteristics.

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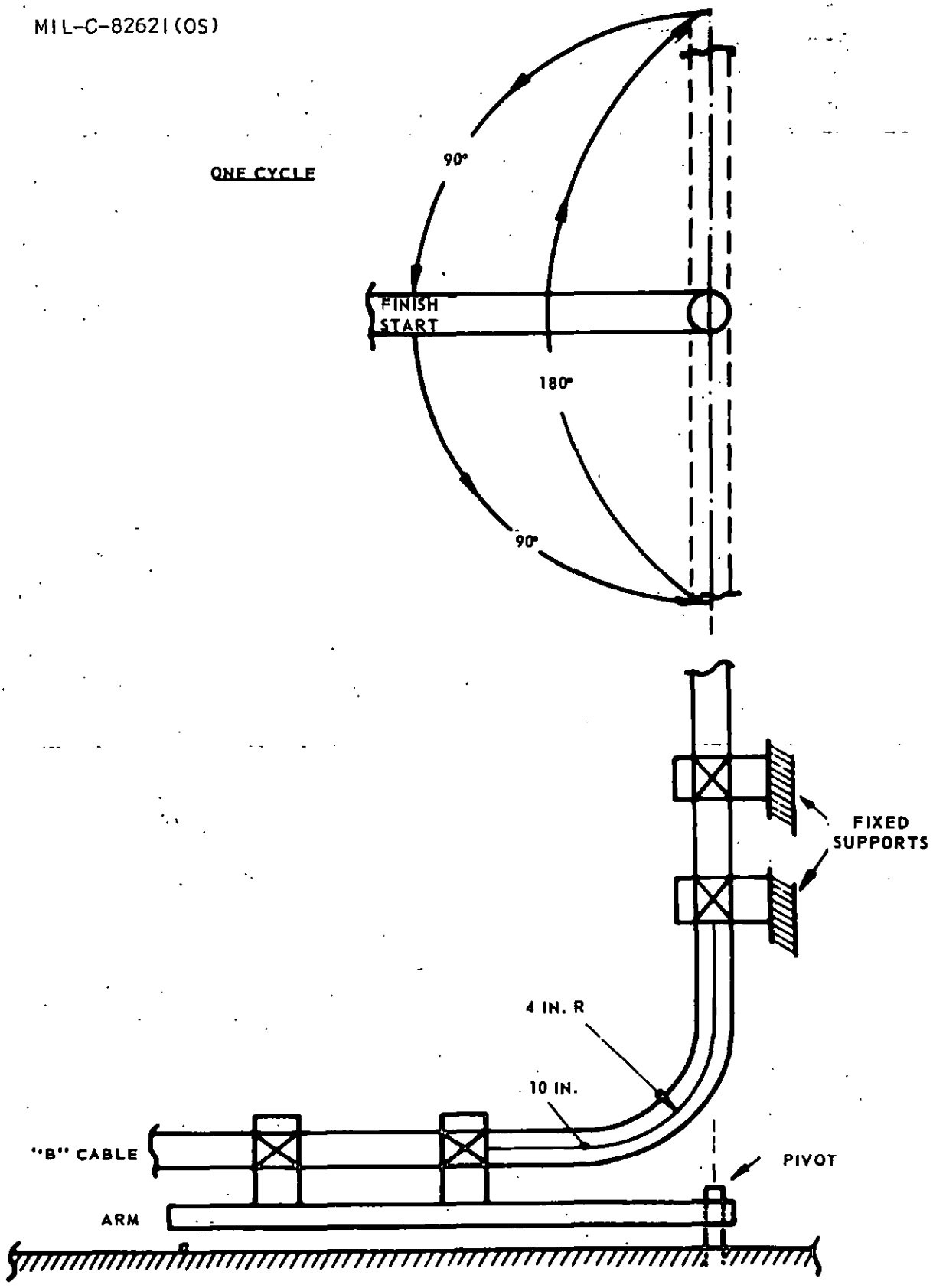


FIGURE 4. TORSION TEST SETUP

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The cable shall then be subjected to a temperature of +122 degrees F for 2 hours. During this 2-hour period the requirements of 3.4.2 shall be met.

(a) After 2 hours, the cable shall be subjected to five 24-hour cycles of 16 hours at +122 degrees F and 8 hours at +86 degrees F. The relative humidity shall be maintained constant at 95 percent during the five cycles. While at +122 degrees F the requirements of 3.4.2 shall be met.

(b) Upon completion of five cycles, the temperature shall be maintained at +122 degrees F for 8 hours. During the 8-hour period, the relative humidity shall be 50 percent for the first 2 hours, 95 percent during the next 4 hours, and 40 percent during the final 2 hours. At 30-minute intervals the continuity of all conductors shall be checked. Upon completion, the requirements of 3.4.2 shall be met.

NOTE: The transitions between temperatures shall be accomplished within the 8-hour period so that the time at the high temperature is approximately 16 hours. Each transition shall not exceed 1½ hours.

4.4.5.3 Temperature and humidity (storage mode). The cable assembly, in the packaged condition, shall be subjected to a 28-day temperature/humidity test. The total test is made up of two 14-day temperature and humidity cycles. The term "chamber temperature," as used in this test, refers to the temperature of the air immediately surrounding the equipment. Chamber temperature may be changed (1) by varying the temperature of a single chamber or (2) by moving the test specimen from one constant temperature chamber to another. The fluctuations from all specified temperatures shall not exceed 10 degrees F. The following test sequence is based on a start at 0800 Monday. Regardless of the actual day and time the test is initiated, there shall be no deviation from the test sequence as prescribed.

(a) At 0800 Monday place the test specimen in a chamber which has been maintained at -65 degrees F for at least 2 hours immediately prior to 0800 (humidity not controlled).

(b) At 1600 Monday the chamber temperature shall be changed to +160 degrees F and 95 +3 -0 percent humidity. The chamber temperature must reach +160 degrees F and 95 percent humidity not later than 1800.

(c) The test specimen shall be held under these conditions until 0800 Tuesday, at which time the temperature decrease shall begin. The rate of decrease of chamber temperature shall be equal to or greater than 36 degrees F per hour for at least 2½ hours. Thus, at 1030, the temperature of the air surrounding the test specimens must be +70 degrees F or lower. The chamber temperature must reach -65 degrees F no later than 1400 and this temperature shall be held until 1600.

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(d) At 1600 Tuesday the chamber temperature shall be changed to +160 degrees F and 95 +3 -0 percent humidity as rapidly as practicable (not later than 1800) and held until 0800 Wednesday.

(e) On Wednesday, Thursday, and Friday the operations performed on Tuesday shall be repeated.

(f) The chamber temperature shall be raised to +160 degrees F and 95 percent humidity on Friday evening and shall be so maintained until 0800 Monday.

(g) At 0800 Monday the sequence of operations described above for Tuesday of the first week shall be followed and shall be repeated daily until Friday of the second week.

(h) On Friday, the chamber temperature shall be reduced to -65 degrees F and this temperature shall be maintained until 0800 Monday of the third week. The cycle will be complete at 0800 Monday.

4.4.5.3.1 The above sequence of temperature and humidity conditions will constitute one cycle. Two such cycles shall be applied. The second cycle shall be completed at 0800 Monday of the fifth week. At that time, the test specimen shall be allowed to return to room temperature. Without drying, wiping, or baking, the test specimen shall meet the requirements of 3.4.1 and 3.4.2.

4.4.5.4 Tropical environment. For this test, the test specimen (a length of completed cable), shall be mounted in the test chamber in such a manner as to prevent exposing the cable ends to the environment.

(a) The test specimen shall be mounted as specified above in a mold chamber in which a temperature of 86 degrees F ±5 degrees F and a relative humidity of 90 percent ±5 percent is maintained. The test specimen, while so housed, shall be sprayed with a suspension of mixed spores. The test specimen shall be maintained in this condition for a period of 28 days. At the end of this period, the test specimen shall be allowed to return to room temperature, and it shall be examined for evidence of being nutrient to fungi. If the test specimen shows susceptibility to fungi, colored photographs shall be taken.

(b) At least five groups of fungi shall be used in the test specimen inoculation. Five such groups are listed in MIL-T-18404 and one type of fungus from each group shall be used. In preparing the spore suspension, distilled water (having a pH value between 5.8 and 7.2 at 77 degrees F) shall be prepared by sterilization in convenient containers (approximately 100 milliliters each). Approximately 10 milliliters of the solution shall

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then be introduced directly into a stock culture and shaken vigorously in order that a well sporulated suspension will result without disturbing the agar.

(c) This process shall be repeated for each type fungus. The separate spore suspension from the various types of fungi shall then be mixed together in an atomizer to provide a composite suspension. Actively growing cultures in ripe fruiting condition shall be used for each inoculation and under no condition will the inoculum be kept for more than an 8-hour period.

(d) The cable sheath shall show no evidence of being nutrient to fungi and the cable shall meet the requirements of 3.4.1 (Dielectric strength) and 3.4.2 (Insulation resistance). The continuity of all conductors shall be maintained.

4.5 Packaging, packing, and marking. Packaging, packing, and marking shall comply with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The cable shall be wound on reels or in coils as specified in MIL-C-915. Cable in lengths of 250 to 1000 feet shall be shipped on reels; whereas cable in lengths of 25 to 249 feet shall be packaged in coils.

5.2 Packing

5.2.1 Reels. Cable wound on reels will require no further packing.

5.2.2 Coils. Cable in coil shall be packed for domestic shipment, if specified in 6.2, in accordance with MIL-C-915.

5.3 Marking. Each reel and coil of cable shall be marked for shipment in accordance with MIL-C-915, with the exception that color stripes shall be as follows for the particular year of manufacture. This color cycle shall repeat itself every fifth year.

Year of Manufacture	Identifying Color
1972	Green
1973	Orange
1974	Blue
1975	White
1976	Red

6. GENERAL

6.1 Intended use. The electric cable described herein forms a part of the SUBROC missile control cables (LD 162872 and LD 162873). These units

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complete the fire control circuit between the torpedo tube breech door and the fire control console.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Whether preproduction sample is required (see 4.2.2.1 and 4.3.3).

(c) Lot acceptance sample, if different from 4.2.2.2.

(d) Packaging and packing required.

Custodian
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